

Homework

Human Reproduction · Year 9 Science

about 80 minutes · A / M / E

Do Part 1 and Part 2 on two different days before the exam (about 40 minutes each). **Print this sheet at 100 %:** the graph grid is true size.

Part 1 · Sex cells, the two systems and the menstrual cycle

1 Write the meaning of each word. **[A]**

(a) gamete

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(b) ovulation

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(c) hormone

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2 Is each part found in males only, in females only, or in both sexes? Write **M**, **F** or **B**. **[A]**

bladder _____ cervix _____ epididymis _____ urethra _____

oviduct _____ prostate gland _____ ovary _____ anus _____

3 As soon as one sperm cell has entered an egg cell, the membrane of the egg changes. **[M]**

(a) State what this change does.

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(b) Explain why this is important for the number of chromosomes in the zygote.

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4 Explain why a girl cannot become pregnant before puberty. **[E]**

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5 State what happens at the menopause, and the age at which it usually happens. **[A]**

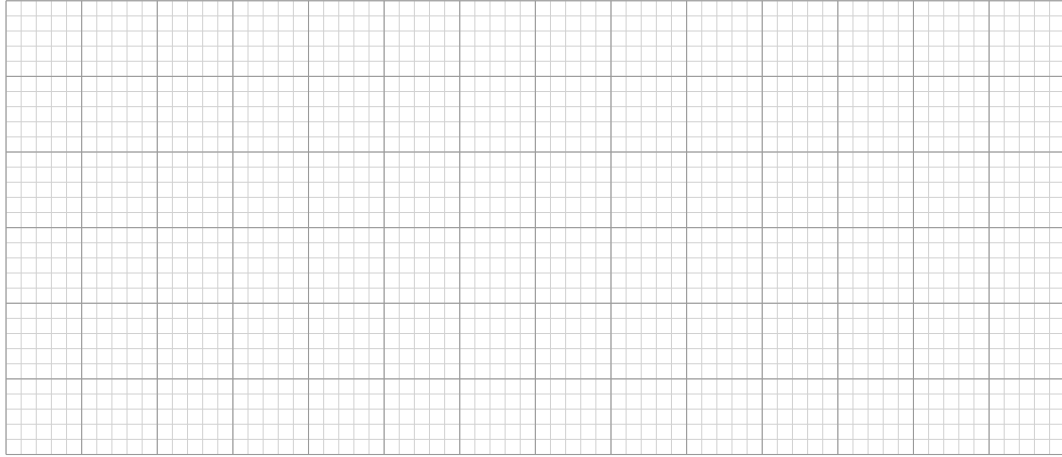
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6 A woman measures her body temperature at the same time every morning during one cycle.

Day of the cycle	2	6	10	13	15	18	22	26
Temperature (°C)	36.4	36.4	36.5	36.4	36.8	36.9	36.9	36.8

- (a) Plot the results on the grid. Label both axes with units, choose an even scale, and join the points with straight lines. **[M]**



- (b) Describe the pattern in the results. **[M]**

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- (c) Use the results to suggest the day of ovulation. Explain your answer. **[E]**

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7 A student is asked to describe the menstrual cycle.

Sam's answer

The menstrual cycle lasts about 14 days. LH makes the lining of the uterus break down. If the egg is fertilised, the woman has a period.

Sam's answer has 3 errors. Find each one and write the correct science. **[M]**

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Part 2 · Fertilisation, chromosomes, pregnancy and fertility

8 Write the meaning of each word.

[A]

(a) zygote

(b) implantation

(c) premature

9 A body cell of a dog has 78 chromosomes.

[A / M]

(a) Number of chromosomes in a dog's sperm cell: _____ (b) in a fertilised dog egg: _____

(c) Explain your answer to (a).

10 Explain why about half of all babies are boys.

[M]

11 Complete the table about twins.

[A]

	Identical twins	Non-identical twins
Number of eggs fertilised	_____	_____
Number of sperm cells needed	_____	_____
Do they have the same genes?	_____	_____
Are they always the same sex?	_____	_____

12 The placenta has many villi and a rich blood supply. Explain how each feature helps the foetus. [E]

13 A pregnant woman slips and falls on her side. Explain why the foetus is usually not hurt. **[M]**

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14 Name the three stages of birth in order. State what leaves the uterus in stage 2 and in stage 3. **[A]**

Stage 1 _____ Stage 2 _____ Stage 3 _____

Leaves in stage 2: _____ Leaves in stage 3: _____

15 The table shows how many babies are born premature in three kinds of pregnancy.

Babies in the pregnancy	one	twins	triplets
Born premature (%)	7	57	93

(a) Describe the pattern in the table. **[M]**

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(b) Suggest why babies from a multiple birth are more likely to be premature. **[E]**

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(c) Use the table to explain why fertility treatment can be a risk for the babies. **[E]**

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16 A student is asked to describe IVF.

Sam's answer

In IVF the baby grows in a test tube until it is ready to be born. IVF is used when a woman's sperm ducts are blocked.

Sam's answer has 2 errors. Find each one and write the correct science. **[M]**

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